

GABRIEL YU

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EDUCATION

National University of Singapore

Jul 2025 - Present

Bachelor of Computing in Computer Engineering

- Relevant course work: Computer Architecture, Embedded Systems, Data Structures & Algorithms, Operating Systems, Digital Design, Software Engineering, Computer Networks, Machine Learning, Databases, and Cybersecurity
- Expected Graduation Date: May 2028

KEY SKILLS

- Frontend: ReactJS, CSS, Bootstrap, Typescript
- Backend: NodeJS, Python Flask
- Database: SQL, DynamoDB
- Cloud and DevOps: AWS (EC2, ECS, SNS, IoT Core, CloudWatch, S3), Docker, Docker Compose, Terraform, Git, GitHub Actions
- Hardware and Embedded Systems: Arduino Uno, ESP8266, ESP32, Raspberry Pi, PIC Microcontroller

EXPERIENCE

Singapore Polytechnic, IoT Engineering Intern

Mar 2024 - Aug 2024

- Re-architected backend from a monolithic system to microservices, improving modularity and scalability of the IoT platform for sensor data collection.
- Orchestrated services with Docker Compose, streamlining local development and multi-service deployments.
- Migrated platform from on-premise to AWS, deploying ReactJS frontend, NodeJS microservices, and MQTT broker on EC2 and later ECS with Application Load Balancer for path-based routing.
- Designed and implemented auto-scaling functionality within AWS, enabling platform to handle up to ~9,000 requests/min while minimising infrastructure costs.
- Developed a data archival workflow in Apache Airflow, automating transfers from MySQL (RDS) to Amazon S3 for long-term storage and cost optimisation.
- Automated cloud infrastructure with Terraform, integrating with GitHub Actions to enable one-click deployment and teardown of full IoT platform, reducing setup time by 300%.

ACHIEVEMENTS

- Lee Hsien Loong Interactive Digital Media Smart Nation Award 2025: Top 25 - Built an AI-based real-time glass abnormality detection system using computer vision (Sobel/Canny preprocessing, single-stage object detection, semi-supervised learning) with AWS SNS integration for live alerts
- CDL Young SDG Leaders Award: Top 10 Finalists - Proposed sustainability solutions and later, with Schneider Electric, designed Scope 3 decarbonization strategies such as joint R&D, bioleaching for e-waste, and collaborative supply chain models
- Director's Honour Roll, Singapore Polytechnic (2024) - Awarded to top 10% of the Cohort for outstanding academic performance

PROJECTS

- Tele-operated Lunar Rescue Robot - Engineered a dual-operator search and rescue platform featuring a distributed TCP architecture across a Raspberry Pi and Arduino Mega, implementing multi-processed SLAM navigation with zero-copy shared memory, Cartesian-approximation macros for 4-DOF arm articulation, and bare-metal interrupt-driven firmware for zero-latency hardware control.
- Safe Pro Max - Created a cloud-connected smart safe with RFID, fingerprint, OTP verification, and TR64 compliance, with real-time monitoring and remote control via a web dashboard using ESP32/ESP8266, Arduino IDE, AWS IoT Core (MQTT), DynamoDB, and AWS SNS.
- Elderly Fall Detection System - Developed a non-intrusive monitoring wearable leveraging Bosch XDK sensors, providing multi-stage fall detection, posture monitoring, location tracking, camera capture, and instant alerts via a centralised dashboard in CONPROSYS gateway.
- Voice Activated Home Automation System - Constructed a smart home simulation on a Raspberry Pi 4B integrating RFID/keypad door lock, scene-based automation, web interface, and Google Voice Recognition API for secure, automated, and voice-controlled household management.
- Healthy Trail (Cycling & Fitness Mobile App) - Created a mobile application with an accompanying website to promote cycling, featuring fitness tracking, goal setting, reward points, and a microcontroller-based IoT cycling tracker with Bluetooth and GPS for real-time distance monitoring.

PROFESSIONAL DEVELOPMENT

Raffles Hall Voices (NUS), Head

Mar 2026 - Present

- Direct timeline planning, organisation-wide communications, and general administrative duties for Raffles Hall Choir.
- Guide overall direction of the CCA, liaising with the Junior Common Room Committee (JCRC) for event execution while prioritising member wellbeing and engagement.

Boy's Brigade Share-a-Gift, Member

Nov 2024 - Dec 2024

- Engaged in door-to-door collection of over 100 food donations from residents across communities, contributing to initiative's goal of bringing cheer to less fortunate people in Singapore.
- Strengthened social skills through meaningful interactions with over 30 diverse residents per group, enhancing ability to connect with individuals from varied backgrounds.
- Collaborated with a group of 6 dynamic members, honing teamwork and interpersonal skills to ensure smooth execution of event.